

Literature Review Draft

A Comparative Analysis of Regression Models for Predicting Life Expectancy Using WHO and Statistics Canada Dataset

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1. Life Expectancy Regression Analysis – Wenshan Zhang (2024)

This study employs multiple linear regression to analyze life expectancy determinants, focusing on vaccination, health status, and socioeconomic conditions. The research applies data preprocessing, assumption checking, and heteroscedasticity corrections, including Weighted Least Squares and the Goldfeld-Quandt test. The study identifies key predictors such as immunization rates, GDP, and mortality. While the approach provides valuable insights, reliance on linear regression limits flexibility. More robust models like XGBoost could improve accuracy. Additionally, stratifying models by development status enhances prediction but lacks validation against external datasets. The study effectively highlights public health implications but could incorporate more advanced machine learning techniques.

2. Life Expectancy Prediction through Analysis of Immunization and HDI Factors using Machine Learning – Lakshmanarao et al. (2022)

This study applies machine learning to predict life expectancy, emphasizing immunization and Human Development Index (HDI) factors. Using WHO data, it tests regression models, including Linear Regression, Support Vector Machines, Decision Trees, and Random Forest, finding Random Forest most accurate ($R^2 = 0.96$). The research effectively demonstrates the impact of immunization on longevity but

lacks an in-depth feature selection discussion. The dataset is comprehensive, yet results may benefit from validation against additional sources. Furthermore, testing ensemble models like XGBoost or deep learning methods could enhance performance. Despite these limitations, the study provides useful insights into healthcare policy and life expectancy modeling.

3. An Application of a Supervised Machine Learning Model for Predicting Life Expectancy – Lipesa et al. (2023)

This study develops a supervised machine learning model for life expectancy prediction, comparing XGBoost, Random Forest, and Artificial Neural Networks (ANN). Using WHO and UN data from 193 countries (2000–2015), it finds XGBoost most effective, achieving an RMSE of 2.402. The research highlights the influence of health, socioeconomic, and behavioral factors. While XGBoost outperforms previous models, the study lacks an exploration of deep learning techniques like LSTM, which could model temporal trends. Additionally, feature interpretability (e.g., SHAP values) would strengthen policy relevance. The findings support XGBoost's efficiency but could benefit from further model refinement and real-world policy applications.

4. Regression Model Selection for Life Expectancy Prediction: A Comparative Analysis of Imputation Techniques – Bahadure et al. (2024)

This study examines life expectancy prediction through various regression models while assessing the impact of different imputation techniques on missing data. It finds that advanced imputation methods significantly improve model accuracy. The research effectively addresses a critical challenge in life expectancy modeling, ensuring reliable predictions despite data gaps. However, it does not explore computational trade-offs between imputation techniques or compare them with deep learning approaches. While methodologically sound, the study could integrate real-world applications, such as healthcare policy recommendations. Its findings are valuable for improving data preprocessing in life expectancy models but would benefit from broader model comparisons.

5. Testing for Linearity in Boundary Regression Models with Application to

Maximal Life Expectancies – Kampf & Meister (2020)

This study explores boundary regression models where errors are always negative, meaning the regression function represents an upper bound rather than an expectation. The authors develop two goodness-of-fit tests to assess whether maximal life expectancy follows a linear trend. Using L^2 -distance-based test statistics, they apply these methods to life expectancy data and find significant deviations from linearity. While the study provides a rigorous statistical framework, its reliance on boundary regression limits applicability to broader demographic trends. Incorporating causal factors and machine learning approaches could enhance interpretability and predictive power. The research is methodologically sound but lacks policy implications.

References

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